

From access to adherence: fostering professional capital and continued engagement in blended teacher workshops

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Abstract

Purpose – As teacher professional development migrates to digital formats, ensuring continuous engagement transcends mere technological adoption. Without sustained participation, online initiatives fail to impact teaching practice or build professional capital. This study investigates the factors driving teachers to continue participating in extensive online professional learning communities, specifically expert teacher workshops in China.

Design/methodology/approach – Viewing these workshops as distinct social ecosystems, this research integrated the technology acceptance model and expectation confirmation theory within a social support framework to model teacher motivation comprehensively. Data from 527 practicing teachers were analyzed using partial least squares structural equation modeling.

Findings – A dual pathway mechanism drives sustained engagement. A rational pathway is anchored in perceived usefulness, while an affective pathway is shaped by overall satisfaction. Multidimensional social support significantly bolsters both routes. Instrumental support lowers technical barriers, whereas informational and emotional support confirm platform value and foster collegiality.

Originality/value – Moving beyond single-theory explanations, this study identifies a socially embedded continuance mechanism that helps explain how digitally mediated participation may contribute to professional capital over time. The model does not directly measure professional capital; rather, it shows how sustained participation, supported by instrumental, informational and emotional support, creates conditions for the development of human, social and decisional capital within expert-led professional communities.

Keywords Professional learning communities, Social support, TAM-ECT, Continuance intention, Expert teacher workshops

Paper type Research article

1. Introduction

The rapid evolution of Fourth Industrial Revolution technologies, encompassing artificial intelligence, big data, and virtual reality, continues to fundamentally reshape the educational ecosystem. These shifts place increasingly rigorous demands on teachers' digital competencies and pedagogical perspectives (Teo *et al.*, 2021). Within the sphere of in-service teacher education, the sudden proliferation of generative AI tools compels educators not merely to master technical innovations rapidly but to immerse themselves in a professional development ecosystem that supports ongoing learning and adaptive resilience (Brandão *et al.*,



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2024). However, the variability in how teachers accept and integrate these new technologies (Chen *et al.*, 2025) creates unprecedented challenges for maintaining continuous professional growth.

Teacher professional development (TPD) is widely recognized as a primary driver for enhancing educational quality and student learning outcomes (Darling-Hammond *et al.*, 2017), serving as a catalyst for broader instructional innovation and the transformation of teaching practices (Berkovich and Becher, 2025). Compared to traditional offline training, which is often constrained by temporal and spatial rigidities, online training offers a more flexible, efficient, and cost-effective avenue for development (McCoy and McNaughton, 2021). Furthermore, it provides the necessary infrastructure to foster the digital skills required for cutting-edge educational technologies, such as generative AI (Lan *et al.*, 2025).

Against this backdrop, Expert Teacher Workshops (ETWs), commonly rendered in English-language scholarship as Master Teacher Studios (e.g., Zheng *et al.*, 2019), have become a distinctive form of blended teacher development in China since the early twenty-first century. Within China's public teacher-development system, ETWs are commonly organized through education authorities, schools, and platform-supported professional networks. Rather than operating as stand-alone short courses, ETWs are policy-supported, cross-school professional learning arrangements organized around a recognized master teacher and sustained through ongoing online and offline activities. In Zhejiang, this model is platform-mediated through *Zhijianghui*, which hosts named-teacher studio functions and related teacher-development applications. Participation is typically organized through role-differentiated arrangements and platform-based entry procedures rather than through open, one-off enrollment. ETWs should therefore be understood less as temporary training events than as institutionally organized infrastructures for sustained teacher learning, expert mentoring, and regional teacher development.

Conceptually, ETWs overlap with professional learning communities (PLCs) and communities of practice (CoPs) because they rely on sustained interaction, shared inquiry, and the improvement of teaching practice (Stoll *et al.*, 2006). They also resemble online teacher professional development programs in that participation is mediated by digital platforms, curated resources, and organized learning tasks. Yet ETWs are not a direct equivalent of any single international form. Their distinctive features lie in named expert leadership, formal role differentiation, cross-school recruitment, and platform-based governance backed by educational authorities. ETWs may therefore be understood as a Chinese, institutionalized variant of online professional learning community design—one that combines vertical mentoring with horizontal peer exchange and connects professional learning to an explicit regional development agenda.

Nevertheless, the transition to online environments is not without friction. Online training frequently presents challenges, including difficulties in technology adaptation, insufficient depth of interaction, and alarmingly low rates of sustained participation (Mulaimović *et al.*, 2025). Research underscores that a teacher's intention to continue with online training is not merely a reflection of their initial evaluation but is a decisive factor in whether professional development programs can achieve a long-term, transformative impact on the profession (Garet *et al.*, 2001).

In the domain of educational technology, most prior studies have focused on the initial adoption of specific technologies by students or teachers (W. Zheng *et al.*, 2024). While international research on online teacher professional development communities, professional learning networks, and networked communities of practice establishes that long-term involvement depends on the quality of interaction, the usability of shared resources, and the strength of collegial exchange (Lantz-Andersson *et al.*, 2018; Lu *et al.*, 2024), the existing literature primarily emphasizes early participation patterns, policy frameworks, or basic platform design (Takizawa *et al.*, 2024), and has often conceptualized professional networks as relatively decentralized or voluntary affiliations. Far fewer studies have examined continuance intention within institutionally organized, expert-led, and role-differentiated blended

communities such as Chinese ETWs, or theorized how structured external social support is translated into teachers' cognitive and affective evaluations over time. To address this gap, this study treats ETWs as socially organized professional learning environments and investigates how sustained engagement is shaped jointly by platform experience and the multidimensional support structures surrounding participation.

To rigorously examine these operational influences, this study introduces Social Support Theory to capture the multifaceted external assistance teachers receive during training. Building upon this foundation, we employ the Technology Acceptance Model (TAM), specifically its constructs of perceived ease of use (PEOU) and perceived usefulness (PU), to understand how such social support translates into rational evaluations of the platform's value (Almenara *et al.*, 2021). However, recognizing that TAM alone may not fully account for the emotional experiences and expectation assessments that emerge during long-term usage (Teo, 2015), we incorporate Expectation Confirmation Theory (ECT). This allows for the integration of satisfaction and continuance intention at the behavioral level, ultimately forming a comprehensive three-tier framework of external support, perception, and behavior.

Guided by this integrated perspective, the study addresses three core questions:

- RQ1. How does social support shape teachers' continuance intention via their perceived experiences of the platform?
- RQ2. What is the relative significance of rational evaluation (e.g. perceived usefulness) versus emotional experience (e.g., satisfaction) in determining the decision to maintain engagement?
- RQ3. Does merging Social Support Theory with the TAM-ECT framework provide a more comprehensive explanation for teachers' continued engagement in Expert Teacher Workshops?

A conceptual clarification is necessary here. Although the title refers to professional capital, the empirical model does not directly measure human, social, or decisional capital as outcomes. Instead, it examines continuance intention as a proximal mechanism through which professional capital may be developed in digitally mediated professional communities. Sustained participation is analytically important because it creates conditions for expert knowledge to circulate, collegial trust to deepen, and professional judgment to be refined through repeated interaction.

By empirically testing a Partial Least Squares Structural Equation Model (PLS-SEM), this research elucidates how instrumental, emotional, and informational support collectively fortify teacher training. We verify the parallel influence of rational evaluation and emotional experience on continued use, demonstrating that teachers exhibit notably stable continuance intentions under the dual influence of perceived usefulness and satisfaction. These findings offer a nuanced model for designing effective and enduring online TPD, highlighting the necessity of cultivating a socially supportive ecosystem that addresses both the practical and emotional needs of the professional community.

2. Theoretical framework and hypotheses development

Against the backdrop of increasingly close integration between digital technologies and educational concepts, teacher professional development has gradually shifted from traditional offline approaches to blended formats that combine real-time interaction with self-directed learning (Lu *et al.*, 2024; Xu *et al.*, 2026). In this context, Expert Teacher Workshops (ETWs) offer a more flexible and personalized environment for professional growth by incorporating expert guidance, online resources, and opportunities for independent study (Liu, 2022). In Zhejiang Province, ETWs are typically organized as provincially or locally recognized workshops led by a master teacher and structured through differentiated roles—such as

workshop leaders, subject leaders, core trainees, and wider network participants. Participation is commonly mediated through platform-based review procedures on systems like Zhijianghui, rather than open, one-off enrollment. This organizational design matters analytically because it makes ETWs simultaneously a policy-supported training arrangement, a professional community, and a platform-mediated network. Ensuring sustained and in-depth teacher engagement in such extensive and cross-regional blended training platforms is therefore crucial for maximizing their impact.

To systematically examine teachers' continuance intention within these blended training platforms and its associated mechanisms, this study draws upon three theoretical perspectives: social support theory, the TAM, and the ECT. These perspectives are combined sequentially rather than additively. First, at the external level, Social Support Theory captures the relational and organizational conditions under which teachers receive instrumental, informational, and emotional assistance from experts, peers, and the broader workshop structure. Second, at the perceptual level, we investigate how TAM concepts (i.e. perceived ease of use and perceived usefulness) shape teachers' evaluations of the platform. In this model, TAM explains how external support is translated into rational appraisals of usability and value. Finally, at the level of continuous motivation, we employ ECT to elucidate the relationships among satisfaction, expectation confirmation, and continuance intention.

ECT is therefore used to explain how post-adoption experience confirms or disconfirms initial expectations and whether those experiences are strong enough to sustain continued engagement. This sequencing allows the integrated model to answer the three research questions more fully than any single theory alone. The subsections below detail the core concepts of social support theory, TAM, and ECT, illustrating their relevance to this study and presenting the hypotheses and structural equation models that guide our empirical work.

2.1 Social support theory

Social Support Theory provides a critical lens for understanding the relational dynamics within online teacher training, moving beyond individual coping mechanisms to the collective construction of professional capital. Cobb (1976) originally conceptualized social support as information leading individuals to believe they are cared for and valued, a notion that remains pivotal in digital environments where isolation can impede learning. In the context of TPD, social support acts as a buffer against the anxieties of technological disruption while simultaneously fueling self-efficacy and resilience (Kärner *et al.*, 2021).

Adopting the framework by Jolly *et al.* (2021), we categorize this support into three distinct dimensions that collectively foster a supportive ecosystem: (1) *Instrumental Support* serves as the logistical foundation, offering technical aid and operational guidance to lower the barriers to digital entry (Zhang *et al.*, 2023). (2) *Informational Support* functions as the intellectual currency of the community, providing instructional feedback and facilitating knowledge co-creation across regional boundaries (Karatay and Hegelheimer, 2021). (3) *Emotional Support* offers the necessary psychological validation through peer recognition and encouragement, crucial for sustaining motivation amidst professional pressures (Peng *et al.*, 2024).

By integrating these dimensions with TAM and ECT, this study treats social support not simply as an external input but as a condition that helps transform a digital platform from a technical tool into a professional learning environment. This framing is consistent with Hargreaves and Fullan's (2012) conception of professional capital as the interrelationship of human, social, and decisional capital. In the present model, instrumental support helps teachers participate with confidence, informational support helps professional knowledge circulate and become usable, and emotional support reinforces the trust and reciprocity needed for collective work. The model therefore does not claim to measure professional capital directly; rather, it examines a participation mechanism through which these forms of capital may gradually be developed over time.

2.2 Technology acceptance model

Davis (1989) proposed the TAM, drawing on the theory of reasoned action, to elucidate the key factors influencing users' acceptance and utilization of new technologies. The model primarily revolves around two constructs: perceived usefulness (PU) and perceived ease of use (PEOU). Perceived usefulness refers to an individual's belief that using a particular technology will enhance work or learning performance, whereas perceived ease of use denotes one's perception of how effortless it is to operate the technology, often indirectly shaped by factors such as social influence (Kemp *et al.*, 2024).

Within online teacher training, if teachers perceive the platform as easy to use and see its genuine value in advancing their professional competencies, they tend to develop favorable attitudes toward the system and exhibit higher intentions to continue using it (Almenara *et al.*, 2021). Importantly, however, technology adoption among teachers is not solely an individual cognitive phenomenon; it is also significantly affected by external social support (Ifinedo *et al.*, 2018).

As shown in Figure 1, instrumental support, information support, and emotional support each enhance teachers' perceptions of platform ease of use and usefulness through distinct channels:

- (1) *Emotional support (ES)* fosters a climate of trust and belonging, thereby reinforcing professional identity and self-efficacy (Pozo-Rico *et al.*, 2023).
- (2) *Information support (INS)* reduces uncertainty about the platform's value by supplying high-quality teaching materials and timely feedback, thereby cultivating more positive expectations of platform functionality (Qin *et al.*, 2017).
- (3) *Instrumental support (IS)* alleviates concerns about technological complexity through technical assistance and strategy training, ultimately boosting digital literacy and promoting sustained platform engagement (Yao and Wang, 2024).

Adopting social support theory as part of TAM's conceptual framework, this study examines how these three forms of support feed into perceptions of ease of use and usefulness, thus shaping teachers' attitudes toward platform utilization and their continuance intention. Based on the above discussion, we propose the following hypotheses:

- H1. Instrumental support positively influences perceived ease of use.
- H2. Emotional support positively influences perceived usefulness.
- H3. Information support positively influences perceived usefulness.

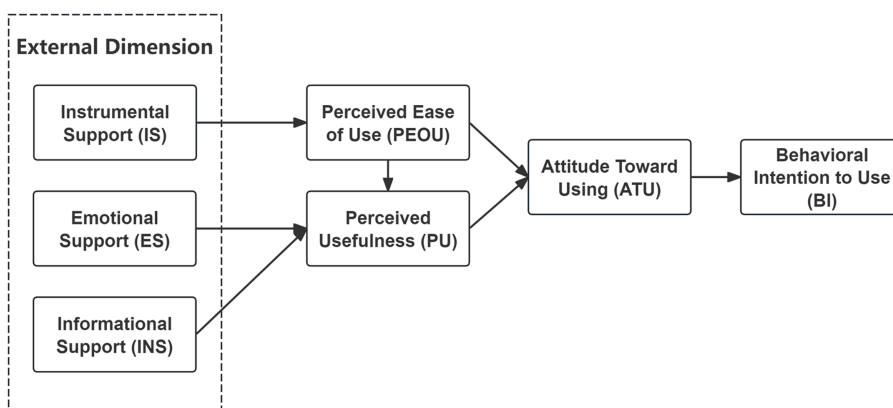


Figure 1. TAM model with external support. Source: Authors' own work

2.3 Expectation confirmation theory

Originally introduced by [Oliver \(1977\)](#) and later extended to the context of information system continuance by [Bhattacharjee \(2001\)](#), the ECT elucidates how the alignment between users’ preliminary expectations and their actual experiences shapes satisfaction and subsequent usage intentions ([Cheng, 2022](#)). In ECT (see [Figure 2](#)), the central constructs include expectation confirmation, perceived usefulness, satisfaction, and continuance intention. When individuals’ real-world interactions with a system meet or exceed their prior expectations, the resulting elevated confirmation fosters higher satisfaction and ongoing motivation to use the system ([Bhattacharjee, 2001](#)).

In the context of online teacher training, teachers often begin with certain assumptions about platform functionality, interaction modes, and developmental benefits. If, during use, they experience meaningful emotional connections and gain access to the high-quality professional resources they anticipated (or better), they are inclined toward greater confirmation and satisfaction ([Pan et al., 2024](#)). Emotional support and information support are particularly salient in shaping both initial expectations and their eventual fulfillment: emotional support helps alleviate teachers’ anxiety regarding novel technologies or instructional approaches, and information support reinforces teachers’ perceptions of the platform’s utility by providing relevant resources and feedback. Accordingly, we hypothesize:

- H4. Emotional support positively influences expectation confirmation.
- H5. Information support positively influences expectation confirmation.

Moreover, perceived usefulness within ECT reflects teachers’ holistic judgment of whether the platform’s functions align with their professional needs—a notion consistent with the TAM constructs of PU and PEOU ([Obeid et al., 2024](#)). When teachers find the platform both easy to operate and effective, they become more likely to confirm their initial expectations and feel satisfied overall. Therefore, we propose that expectation confirmation, perceived ease of use, and perceived usefulness all serve as positive drivers of satisfaction:

- H6. Expectation confirmation positively influences perceived usefulness.
- H7. Expectation confirmation positively influences satisfaction.
- H8. Perceived ease of use positively influences satisfaction.
- H9. Perceived usefulness positively influences satisfaction.

Within the ECT framework, both expectation confirmation and perceived usefulness exert influence on continuance intention through the mediating role of satisfaction ([Han and Sa, 2022](#)). At the same time, perceived usefulness exerts a direct effect on continuance intention, reflecting teachers’ value-based assessment. As teachers’ comprehensive evaluation of the platform and their affective responses (satisfaction)



Figure 2. ECT original model. Source: Authors’ own work

improve, they develop a stronger inclination to keep using it, fostering a reinforcing loop of continued engagement. Consequently, we posit:

H10. Perceived usefulness positively influences continuance intention.

H11. Satisfaction positively influences continuance intention.

Collectively, as shown in [Figure 3](#), the integration of ECT with TAM and social support theory enables a more nuanced exploration of teachers' behavioral patterns in blended training environments. By analyzing how expectation confirmation, perceived usefulness, and satisfaction interact, we can better understand the mechanisms underpinning teachers' decisions to continue or discontinue online training, thereby laying a robust theoretical foundation for subsequent empirical testing.

Furthermore, we propose that each construct's effect on continuance intention may be mediated by specific intervening mechanisms. Drawing on the established direct relationships, we highlight three mediating pathways within the integrated TAM-ECT framework. First, perceived usefulness is posited to mediate the relationship between perceived ease of use and satisfaction (PEOU → PU → SAT), suggesting that perceptions of ease of use enhance satisfaction by augmenting functional value. Second, the influence of expectation confirmation on satisfaction may be partially mediated by perceived usefulness (EC → PU → SAT); when actual experience meets or surpasses initial expectations, it not only contributes directly to satisfaction but also promotes a more favorable reassessment of the platform's practical benefits, thereby strengthening satisfaction. Finally, satisfaction is theorized to mediate the link between perceived usefulness and continuance intention (PU → SAT → CU); while rational judgments about a platform's value can directly encourage continued usage, these assessments also indirectly foster continuance by increasing emotional satisfaction ([Koutromanos et al., 2024](#)). Thus, we present the following mediating hypotheses:

H12. Perceived usefulness mediates the path from perceived ease of use to satisfaction.

H13. Perceived usefulness mediates the path from expectation confirmation to satisfaction.

H14. Satisfaction mediates the path from perceived usefulness to continuance intention.

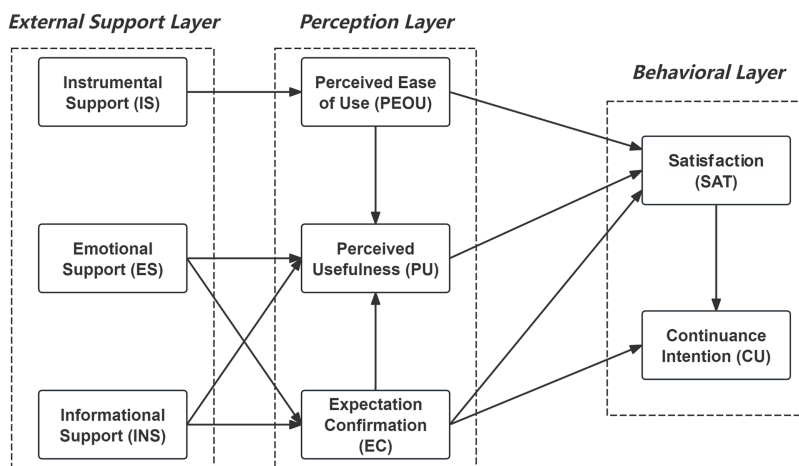


Figure 3. Research theoretical levels model. Source: Authors' own work

3. Methodology and data collection

3.1 Measurement instrument development

This study developed a multidimensional measurement framework by combining social support theory with core constructs from the TAM and ECT. First, the research team adapted existing scales for PU, PEOU, EC, SAT, and CU from [Park \(2020\)](#), [So and Brush \(2008\)](#), and [Dai et al. \(2020\)](#). Subsequently, three dimensions of social support—IS, ES, and INS—were incorporated to capture the external support teachers receive during professional development ([Malecki et al., 2018](#); [Mathieu et al., 2019](#)).

To ensure measurement equivalence across different linguistic contexts, the questionnaire was prepared in Chinese because the participants were Chinese in-service teachers and Chinese was the working language of the workshops. Items adapted from English-language scales were translated into Chinese by the research team and then checked using the back-translation procedure recommended by [Ozolins et al. \(2020\)](#), verifying each item's semantic consistency through a “source language–target language–source language” cycle. Next, a focus group of educational technology experts, workshop facilitators, and participating teachers ($n = 6$) reviewed the initial 31 items, suggesting adjustments to enhance clarity and contextual fit. Following these refinements, a small-scale pilot study ($n = 45$) was conducted to verify scale reliability, validity, and item performance. One item was removed due to a factor loading under 0.60, resulting in a final instrument comprising 8 latent variables and 30 indicators. The instrument employed five-point Likert scales (1 = “strongly disagree,” 5 = “strongly agree”) and included control variables, such as demographic information (gender, age, education level, teaching experience) and workshop participation characteristics (role type, weekly usage time), at the end of the questionnaire.

3.2 Data collection

The study targeted in-service teachers from Expert Teacher Workshops in Zhejiang Province, China. To optimize the instrument and strengthen response quality, a two-stage questionnaire process was used. In the first stage, 45 workshop teachers participated in a pilot test assessing item content, expression, and psychometric properties, followed by minor word choice adjustments. After these revisions, the research team formally distributed the questionnaire via the Wenjuanxing platform. Consistent with [Singer and Ye's \(2013\)](#) guidance on incentivizing survey participation, each teacher received 5 RMB (around 0.70 USD) via WeChat Pay after consenting to participate. This approach yielded 557 initial responses over 14 days.

To ensure data integrity, a two-step standardized screening procedure was adopted. First, the team used Temporal Threshold Filtering to discard 18 surveys completed in under 100 s. Second, an attention-check item (requiring participants to select the option “1”) excluded an additional 12 surveys. Ultimately, 527 valid questionnaires remained (a 94.61% effective response rate). The sample-to-indicator ratio was about 18:1, meeting established criteria for structural equation modeling and ensuring adequate statistical power for subsequent analyses.

3.3 Descriptive statistics

Using the final 527 valid responses, descriptive analysis showed that the sample covered a wide range of professional profiles and participation roles in Expert Teacher Workshops. As shown in [Table 1](#), gender was relatively balanced (47.40% male, 52.60% female), and most respondents were aged 31–40 (62.05%). The majority held a bachelor's degree or below (84.82%), had 10–20 years of teaching experience (57.90%), and held intermediate or senior professional titles (87.48% combined). In terms of studio role, network trainees constituted the largest group (52.94%), followed by core trainees (21.82%), subject leaders (12.71%), workshop leaders (5.12%), and other members (7.40%). Weekly online engagement was generally modest, with 56.93% reporting fewer than three hours and only 9.1% reporting more than six hours per week. Although the study does not claim province-wide statistical representativeness, these characteristics capture substantial heterogeneity. The sample

Table 1. Demographic information of participants ($N = 527$)

Variable	Category	Count	Percentage (%)
Gender	Male	250	47.40
	Female	277	52.60
Age Group	20–30 years	35	6.64
	31–40 years	327	62.05
	41–50 years	147	27.90
	51 years or older	18	3.40
Education	Bachelor's degree or below	447	84.82
	Master's degree or above	80	15.18
Teaching Experience	Less than 10 years	118	22.40
	10–20 years	305	57.90
	20–30 years	86	16.30
	30 years or more	18	3.40
Professional Title	Junior	66	12.52
	Intermediate	243	46.11
	Senior	218	41.37
Role in the Studio	Famous Teacher (Leader)	27	5.12
	Subject Leader	67	12.71
	Core Trainee	115	21.82
	Network Trainee	279	52.94
	Others	39	7.40
Weekly Engagement Time	<3 h	300	56.93
	3–6 h	179	33.98
	6–9 h	32	6.07
	≥9 h	16	3.04

Note(s): Percentages may not total 100% due to rounding

Source(s): Authors' own work

represents a broad cross-section of active ETW participants rather than a single narrow subgroup, while still warranting caution in generalizing beyond Zhejiang.

Before constructing the structural model, extensive scale validation procedures were performed. Mean item scores (1–5 scale) ranged from 3.43 to 3.90 (SD 0.78–0.94), with coefficients of variation below 27.5%, indicating concentrated responses without extreme outliers. Skewness (0.02–0.68) and kurtosis (0.11–1.54) values were under [Kline's \(2023\)](#) recommended cutoffs ($|\text{skewness}| < 2$, $|\text{kurtosis}| < 7$), implying mild deviations from normality. Shapiro–Wilk tests showed p -values below 0.05 for most variables, rejecting the hypothesis of a normal distribution ([Ghasemi and Zahediasl, 2012](#)). Following [Hair *et al.*'s \(2011\)](#) view that Partial Least Squares Structural Equation Modeling (PLS-SEM) is appropriate for such non-normal data, the study employed PLS-SEM to minimize potential biases that can arise with Covariance-Based Structural Equation Modeling (CB-SEM) under non-normal conditions, thus providing robust empirical support for the path analyses.

3.4 Post-survey qualitative follow-up

To contextualize the quantitative findings, we incorporated a concise post-survey qualitative follow-up within an explanatory sequential design. All interviewees were drawn from the survey sample and selected purposively by workshop role to capture perspectives from leadership, core participation, and wider network participation (2 workshop leaders, 2 core members, and 3 network participants). Semi-structured telephone interviews focused on participants' experiences of platform support, usability, expectation confirmation, satisfaction, and reasons for continued participation. Each interview lasted approximately 20 min. The interview material was coded to identify recurrent explanations for the significant

4. Results

Following the PLS-SEM reporting guidelines proposed by Hair *et al.* (2019), this study conducted a hierarchical evaluation of both measurement and structural models. First, the reliability and validity of the measurement model were examined to ensure scale quality and effectiveness. Subsequently, predictive power and robustness indicators were reported at the structural model level, followed by mediation effect analysis to provide a more comprehensive explanation of the influence pathways for teachers' continuance intention.

4.1 Measurement model assessment

4.1.1 Reflective measurement model. This study employed PLS-SEM to investigate teachers' continuance intention in Expert Teacher Workshops. This approach offers advantages in predicting outcomes within complex online learning environments (Shi *et al.*, 2024). To confirm that the measurement model objectively reflected the theoretical constructs, indicator loadings, internal consistency reliability, convergent validity, and discriminant validity were systematically examined.

As shown in Table 2, all latent variables demonstrated composite reliability (CR = 0.871–0.960) and Cronbach's α (α = 0.864–0.960) values above the recommended threshold of 0.70 (Hair *et al.*, 2021; Li *et al.*, 2024a). The initial internal consistency of Perceived Ease of Use (PEOU) was particularly high (CR = 0.960, α = 0.960), suggesting potential indicator redundancy. All observed indicators had factor loadings (0.717–0.945) exceeding the baseline of 0.70, and the Average Variance Extracted (AVE = 0.630–0.857) generally surpassed the threshold of 0.50, indicating good convergent validity (Fornell and Larcker, 1981). It should be noted that the initial CR value for PEOU was 0.960, suggesting potential indicator redundancy (Cepeda *et al.*, 2024). Variance Inflation Factor (VIF) analysis revealed that PEOU4 had a VIF value of 5.684, exceeding the warning standard of 5. After removing PEOU4, the model indicators for PEOU were optimized (CR = 0.947, AVE = 0.866, α = 0.946), effectively reducing the risk of multicollinearity.

Regarding discriminant validity, as shown in Table 3, all constructs had Heterotrait-Monotrait (HTMT) ratios below 0.85 (Hair *et al.*, 2019), and the confidence intervals obtained through bias-corrected bootstrapping did not include 1, further confirming discriminant validity between constructs. Additionally, based on the Fornell–Larcker criterion (Table 4), the square root of AVE for each latent variable was greater than its correlation coefficients with other latent variables, providing further evidence of good discriminant validity. Hence, the reflective measurement model exhibited sufficient reliability and validity, establishing a solid foundation for subsequent structural model analysis.

4.1.2 Collinearity assessment. To assess potential collinearity in the structural model, this study examined the inner VIF values for all predictor paths (see Table 5). All VIF values ranged from 1.000 to 2.364, below the recommended threshold of 3.0, indicating that multicollinearity did not introduce significant bias in parameter estimation (Hair *et al.*, 2019).

4.2 Structural model assessment

In assessing the structural model, this study used the Bootstrapping procedure (5,000 resamples) in SmartPLS 4 to test path relationships and predictive power. Since PLS-SEM inherently emphasizes prediction, traditional goodness-of-fit indicators (such as GoF, SRMR) are more suitable for confirmatory research. Therefore, the focus here was on predictive indicators such as R^2 , Q^2 , and path coefficient effect sizes (Hair and Alamer, 2022; Hair *et al.*, 2021).

Table 2. Summary of the results for the measurement model

Latent variable	Item	Internal consistency Composite reliability >0.70	Cronbach's alpha >0.70	Convergent validity Loadings >0.70	AVE >0.50
Instrumental support	IS1	0.881	0.883	0.843	0.711
	IS2			0.874	
	IS3			0.820	
Information Support	INS1	0.871	0.869	0.853	0.630
	INS2			0.744	
	INS3			0.717	
	INS4			0.852	
Emotional Support	ES1	0.885	0.929	0.825	0.657
	ES2			0.813	
	ES3			0.808	
	ES4			0.797	
Perceived Ease of Use	PEOU1	0.960	0.960	0.923	0.857
	PEOU2			0.923	
	PEOU3			0.929	
	PEOU4			0.945	
Perceived Usefulness	PU1	0.944	0.864	0.919	0.808
	PU2			0.885	
	PU3			0.890	
	PU4			0.910	
Expectation Confirmation	EC1	0.934	0.893	0.879	0.824
	EC2			0.927	
	EC3			0.917	
Satisfaction	SAT1	0.949	0.929	0.911	0.825
	SAT2			0.903	
	SAT3			0.908	
	SAT4			0.909	
Continued Use Intention	CU1	0.939	0.892	0.889	0.796
	CU2			0.890	
	CU3			0.908	
	CU4			0.881	

Source(s): Authors' own work**Table 3.** HTMT values and discriminant validity analysis of constructs

	IS	INS	ES	PEOU	PU	EC	SAT	CU
IS								
INS	0.699							
ES	0.630	0.689						
PEOU	0.655	0.677	0.490					
PU	0.699	0.748	0.673	0.675				
EC	0.630	0.655	0.599	0.630	0.749			
SAT	0.705	0.709	0.627	0.602	0.752	0.696		
CU	0.678	0.691	0.616	0.65	0.725	0.711	0.697	

Note(s) All HTMT values' confidence intervals do not include 1**Source(s):** Authors' own work

4.2.1 Predictive power and robustness. Following the evaluation guidelines suggested by [Hair et al. \(2019\)](#), all endogenous variables achieved substantial or higher explanatory power ($R^2 > 0.360$): PEOU at 0.366, EC at 0.436, PU at 0.647, SAT at 0.554, and CU at 0.522.

Table 4. Correlation matrix with AVE

	AVE	IS	INS	ES	PEOU	PU	EC	SAT	CU
IS	0.807	0.898							
INS	0.805	0.629	0.897						
ES	0.769	0.561	0.628	0.877					
PEOU	0.866	0.605	0.639	0.458	0.931				
PU	0.833	0.634	0.693	0.617	0.639	0.913			
EC	0.823	0.580	0.625	0.558	0.604	0.710	0.907		
SAT	0.825	0.638	0.656	0.574	0.569	0.702	0.659	0.908	
CU	0.845	0.616	0.643	0.566	0.616	0.679	0.675	0.652	0.919

Note(s) Lower triangle shows the correlation coefficients
The diagonal values show the square root of AVE
The AVE values in Table 4 are based on the refined measurement model after excluding PEOU4. Table 2 reports the initial item-level measurement results before this exclusion
Source(s): Authors' own work

Table 5. Inner VIF for the predictors

	IS	INS	ES	PEOU	PU	EC	SAT	CU
IS				1.000				
INS					1.783	1.650		
ES					2.364	1.650		
PEOU					1.912		1.833	
PU							2.349	1.971
EC					1.989		2.186	
SAT								1.971
CU								

Note(s): VIF values from row to column variables
Source(s): Authors' own work

Notably, PU had the highest R^2 of 0.647 (see Table 6) suggesting that multiple paths in the integrated model comprehensively influenced teachers' functional perceptions. Furthermore, the Stone–Geisser Q^2 values (0.315–0.533), calculated via blindfolding, all significantly exceeded zero. With most endogenous variables showing Q^2 values above 0.35, the structural model demonstrated strong predictive relevance and excellent out-of-sample generalization capability (Cohen, 2013).

Table 6. R^2 and Q^2 values of latent variables

Latent variable	Q^2 predict	R^2
Instrumental support	–	–
Emotional Support	–	–
Information Support	–	–
Perceived Ease of Use	0.315	0.366
Perceived Usefulness	0.533	0.647
Expectation Confirmation	0.403	0.436
Satisfaction	0.452	0.554
Continuance Intention	0.436	0.522

Source(s): Authors' own work

4.2.2 Path coefficient analysis. After the overall structural model assessment, the path coefficients and their effect sizes were examined to confirm the strength and substantive contribution of exogenous variables to endogenous variables (see Table 7). Following Hu *et al.* (2025), this study considered f^2 values of 0.02, 0.15, and 0.35 as small, medium, and large effect sizes, respectively. All hypotheses (H1 to H11) were supported at the 0.05 significance level, and the distribution of effect sizes indicated substantial theoretical and practical significance.

From a latent variable perspective, Instrumental Support (IS) had the most pronounced influence on PEOU ($\beta = 0.605$, $f^2 = 0.578$). This path showed a large effect size, partly because only IS influenced PEOU in the model, resulting in relatively concentrated variance explanation. This finding aligns with Skulmowski and Xu's (2022) research, which noted that intuitive operations and task-specific interface designs in digital learning environments can effectively reduce cognitive load and enhance users' overall perception of system usability.

PU, a core variable in TAM theory, was influenced simultaneously by Expectation Confirmation (EC), Emotional Support (ES), and Information Support (INS) in the integrated model, forming a more complex path. Among these, EC had the strongest influence on PU ($\beta = 0.336$, $f^2 = 0.161$, $p < 0.001$), reaching a medium effect size; INS

Table 7. Results of significance tests of the path coefficients

Hypothesis	Relationship	Coefficient (β)	Std	t	95% CI	p	f-square
H1	IS $\rightarrow$ PEOU	0.605	0.037	16.323	[0.531, 0.677]	***	0.578
H2	ES $\rightarrow$ PU	0.190	0.053	3.563	[0.093, 0.300]	***	0.057
H3	INS $\rightarrow$ PU	0.238	0.056	4.248	[0.132, 0.348]	***	0.068
H4	ES $\rightarrow$ EC	0.274	0.062	4.439	[0.157, 0.397]	***	0.081
H5	INS $\rightarrow$ EC	0.453	0.063	7.200	[0.330, 0.576]	***	0.221
H6	EC $\rightarrow$ PU	0.336	0.049	6.803	[0.241, 0.434]	***	0.161
H7	EC $\rightarrow$ SAT	0.283	0.064	4.433	[0.162, 0.410]	***	0.082
H8	PEOU $\rightarrow$ SAT	0.132	0.052	2.531	[0.034, 0.237]	0.011	0.021
H9	PU $\rightarrow$ SAT	0.416	0.074	5.623	[0.274, 0.564]	***	0.165
H10	PU $\rightarrow$ CU	0.436	0.065	6.718	[0.307, 0.561]	***	0.228
H11	SAT $\rightarrow$ CU	0.346	0.062	5.582	[0.243, 0.486]	***	0.162

Note(s): *** $p < 0.001$

Source(s): Authors' own work

Table 8. Qualitative explanations for key structural paths

Structural path	Qualitative theme	Representative quote
IS $\rightarrow$ PEOU	Alleviation of technological anxiety	"Whenever I struggled with uploading my video lessons, the workshop's technical assistant provided step-by-step help. It made the platform feel effortless to use." (Network Participant, 42)
INS $\rightarrow$ PU	Immediate pedagogical transferability	"The lesson plans and expert feedback I received were exactly what I needed for my next day's class. This practical relevance proved the platform's worth." (Core Member, 35)
ES $\rightarrow$ EC	Affective validation and belonging	"Getting encouraged by a master teacher made me feel seen. It confirmed my hope that this wasn't just a resource hub, but a real, supportive community." (Network Participant, 29)
SAT $\rightarrow$ CU	Emotional attachment and habituation	"I am highly satisfied because it feels like a professional home. Logging in every week has become a natural habit rather than a forced administrative task." (Workshop Leader, 48)

Source(s): Authors' own work

($\beta = 0.238, f^2 = 0.068, p < 0.001$) and ES ($\beta = 0.190, f^2 = 0.057, p < 0.001$) both showed small effect sizes. These findings align with [Xianhan et al. \(2022\)](#), who observed that teachers focus more on whether actual experiences match their initial expectations rather than depending solely on external support. The R^2 value of PU reached 0.647, the highest in the model, indicating that multiple paths comprehensively explained teachers' cognitive processes regarding platform functionality.

Satisfaction (SAT), as a comprehensive evaluation of individual perception and expectation fulfillment, was primarily influenced by PU, EC, and PEOU. PU had the most significant impact on SAT ($\beta = 0.416, f^2 = 0.165, p < 0.001$), indicating a medium effect size; EC followed ($\beta = 0.283, f^2 = 0.082, p < 0.001$), showing a small-to-medium effect; while PEOU exhibited a relatively weaker effect ($\beta = 0.132, f^2 = 0.021, p = 0.011$), corresponding to a small effect size. These results echo [Han and Sa's \(2022\)](#) study, suggesting that in teacher professional development contexts, the platform's functional value and fulfillment of expectations play more crucial roles in shaping satisfaction, whereas ease of use, though necessary, is less decisive. The R^2 value for SAT was 0.554, indicating that the model explained about 55.4% of the variance in satisfaction, thereby playing an important mediating role in revealing how continuance intention forms.

Continuance Intention (CU), the ultimate outcome variable, was influenced by both PU and SAT. The direct effect of PU on CU ($\beta = 0.436, f^2 = 0.228, p < 0.001$) was slightly stronger than that of SAT on CU ($\beta = 0.346, f^2 = 0.162, p < 0.001$), and both had medium effect sizes. This is consistent with [Salmerón et al. \(2020\)](#), who argued that in online learning environments, users' rational assessments of a system's value have a more direct impact on usage decisions, though emotional experiences (such as satisfaction) also remain vital. The R^2 value for CU was 0.522, indicating that over half of the variance in teachers' continuance intention could be explained, which is notably persuasive in social science research and demonstrates the integrated model's effectiveness in predicting teachers' continued use of online professional development.

4.2.3 Mediation path verification. To clarify the model's internal mechanisms, three theoretically meaningful indirect paths were tested. Following [Hair et al. \(2014\)](#), this study used Variance Accounted For (VAF) values to assess mediation effects: $VAF > 80\%$ indicates full mediation, $20\% \leq VAF \leq 80\%$ indicates partial mediation, and $VAF < 20\%$ indicates no mediation. The results showed that in the PEOU→PU→SAT path, the indirect effect coefficient was $\beta = 0.082$ ($p = 0.001$) with $VAF = 0.383$, indicating a significant partial mediating role of PU between PEOU and SAT. In the EC→PU→SAT path, the indirect effect coefficient was $\beta = 0.140$ ($p < 0.001$) with $VAF = 0.331$, again reflecting a strong partial mediation effect. Finally, in the PU→SAT→CU path, the indirect effect coefficient was $\beta = 0.144$ ($p < 0.001$) with $VAF = 0.249$, demonstrating another robust partial mediation effect. These findings confirm the core hypotheses of this study: teachers' rational cognition and emotional experience jointly shape their decisions to continue using online training platforms, with rational cognition generally holding a more prominent role, while emotional factors play a key complementary function.

In summary, as illustrated in [Figure 4](#), the structural model integrates the main variables and pathways from social support theory, the technology acceptance model, and the expectation confirmation model. All paths reached significance ($p < 0.05$), further validating the TAM-ECT integrated model grounded in social support for explaining teachers' continuance intention in online professional development. This analysis underscores how both rational and emotional drivers are essential in digital teacher training, while also offering theoretical and practical implications for further investigations into the complex relationships among latent variables.

5. Discussion and implications

5.1 Qualitative insights into path significance

While the PLS-SEM results confirm the structural relationships, the post-survey interviews provide concise explanatory evidence for why these paths emerged in practice (see [Table 8](#)).

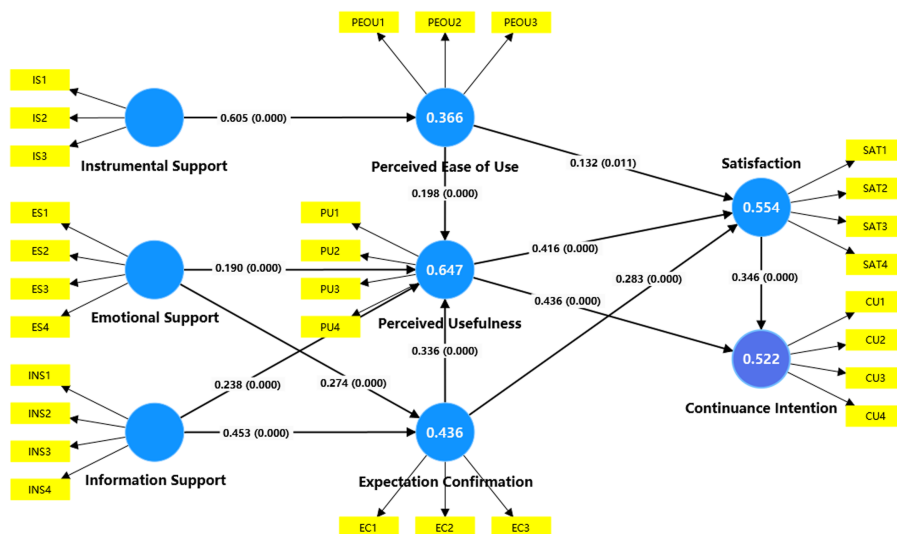


Figure 4. Teachers' continuance intention structural equation model. Source: Authors' own work

The interview material was coded in relation to the significant structural paths and then grouped into higher-order themes that captured recurrent teacher explanations. Table 8 therefore reports path-linked themes rather than stand-alone qualitative categories. For example, instrumental support was repeatedly associated with the alleviation of technological anxiety, information support with the immediate pedagogical transferability of resources and feedback, emotional support with affective validation and belonging, and satisfaction with emotional attachment to continued participation. These qualitative insights do not alter the model, but they make the statistical relationships more interpretable from teachers' lived experience.

5.2 The architecture of engagement: the triadic role of social support

Our findings delineate a hierarchical structure of support required to sustain online professional communities. Rather than operating in isolation, the three forms of social support create a compounded effect that bridges the gap between mere access and deep adherence. First, Instrumental Support acts as the gateway enabler. By significantly influencing perceived ease of use, it reduces the cognitive load associated with technical friction, ensuring that the technology itself does not become an impediment to professional exchange (Skulmowski and Xu, 2022).

Once access is secured, Informational Support serves as the value anchor. It directly enhances teachers' rational assessment of the platform's utility and confirms their professional expectations. This aligns with the view that for knowledge workers, the tangible exchange of pedagogical strategies and feedback constitutes the core value of any professional network (Karatay and Hegelheimer, 2021). Finally, Emotional Support functions as the relational sustenance mechanism. While its direct impact on perceived usefulness is more subtle, its strong connection to expectation confirmation suggests that the sense of belonging and peer validation is essential for converting a transactional training experience into a transformative professional journey (Darling-Hammond *et al.*, 2017; Johnsen, 2001). Collectively, these findings imply that an effective online TPD ecosystem must be architected to simultaneously remove technical barriers, provide rich professional content, and nurture a supportive community culture.

5.3 Dual-track drivers of rational evaluation and emotional experience in online teacher training

Structural equation analysis reveals that teachers' continuance intention in online training is driven by two parallel pathways: the rational evaluation pathway represented by perceived usefulness (PU) and the emotional experience pathway centered on satisfaction (SAT). Specifically, the direct effect of perceived usefulness on continuance intention ($\beta = 0.436$, $f^2 = 0.228$) is slightly stronger than that of satisfaction ($\beta = 0.346$, $f^2 = 0.162$), and perceived usefulness also exerts a partial mediating effect on continuance intention through satisfaction ($\beta = 0.144$, VAF = 0.249). This parallel influence of rational and emotional factors is consistent with the technology adoption model proposed by Bettiga and Lamberti (2017) and supports Li *et al.* (2024b) finding that knowledge-based professionals prioritize the platform's knowledge value.

The integrated TAM and ECT model offers a targeted framework for actionable interventions in online teacher training, suggesting that administrators should design a layered community architecture aligned with specific process needs rather than merely mandating participation (Xu *et al.*, 2025). First, developers must lower initial access barriers through technical onboarding and just-in-time support mechanisms, including interface simplification and real-time help desks. These provisions significantly enhance perceived ease of use ($\beta = 0.605$, IS $\rightarrow$ PEOU), particularly for educators navigating digital literacy challenges (Lane and Lyle, 2011). Second, facilitators can strengthen the platform's functional value by establishing recurring pedagogical routines. Integrating curated resource pathways and expert feedback cycles delivers the high-quality informational support ($\beta = 0.238$, INS $\rightarrow$ PU) necessary to transform static repositories into interactive professional communities (Lantz-Andersson *et al.*, 2018). Third, sustaining affective engagement demands role-based mentoring and differentiated participation pathways. Tailoring recognition and encouragement to specific roles, such as distinguishing between core trainees and wider network participants, provides essential emotional support ($\beta = 0.274$, ES $\rightarrow$ EC) that realistically confirms diverse professional expectations ($\beta = 0.336$, EC $\rightarrow$ PU). Ultimately, while derived from Chinese ETWs, this multifaceted alignment of technical assistance, informational routines, and role-based emotional support provides potentially transferable design principles for comparable expert-led, platform-mediated professional development settings, while remaining grounded in the Chinese ETW context.

5.4 Theoretical contributions: from continuance mechanisms to conditions for professional capital

This study extends traditional technology acceptance research by situating teachers' sustained engagement within the collaborative fabric of the teaching profession. While the classic TAM framework robustly explains individual cognitive processing, it is less well suited to capturing the relational dynamics inherent in organized professional learning environments (Mogaji *et al.*, 2024). By synthesizing Social Support Theory with TAM and ECT, our integrated model shows that continuance intention is shaped not only by functional utility but also by the social conditions of participation within professional networks. This, in turn, clarifies the study's bounded contribution to the literature on professional capital. Following Hargreaves and Fullan (2012), professional capital arises through the interaction of human, social, and decisional capital rather than from isolated individual competence alone. From this perspective, our findings do not demonstrate professional capital directly; rather, they suggest that ETWs may create conditions supportive of its development by reducing participation barriers, facilitating professional knowledge exchange, and sustaining collegial trust over time. Recent work on organizational social capital further reinforces the importance of relational trust, shared understandings, and institutional memory for collective problem-solving and instructional improvement (Jabbar and Holme, 2026), but these are treated here as supportive conditions rather than directly observed outcomes.

The high explanatory power of the model (Continuance Intention $R^2 = 0.522$) validates the premise that “trust” and “community” are as critical as “interface” and “content” in digital learning environments (AlSokkar *et al.*, 2024; Tawafak *et al.*, 2023). Accordingly, the study does not claim to measure professional capital directly. Rather, it identifies continuance intention as a theoretically important bridge between platform participation and the longer-term accumulation of professional capital within expert-led professional communities. This contributes to the literature on professional capital by quantifying how external support networks—comprising peers, mentors, and administrators—can be operationalized to drive behavioral persistence in online settings. Over time, such persistence increases the likelihood that expertise will circulate, collegial relations will deepen, and professional judgment will be refined through repeated interaction. Future research can build upon this “socially embedded” perspective to explore how specific community structures, such as interdisciplinary alliances or AI-augmented tutoring groups, further modulate the relationship between professional interaction and continuous learning.

6. Conclusion and limitations

This study investigated teachers’ continuance intention in online training contexts by integrating TAM and ECT within a social support framework. The findings show that instrumental, informational, and emotional support jointly shape teachers’ cognitive and affective evaluations, thereby encouraging continued participation in Expert Teacher Workshops. Perceived usefulness occupied the core position in teachers’ rational evaluation of platform value, while satisfaction provided affective reinforcement; together, they formed a dual pathway to continuance intention. In substantive terms, these findings do not demonstrate professional capital directly. Rather, they indicate that socially supported and sustained participation in ETWs may contribute to professional capital by creating conditions for professional knowledge sharing, collegial trust, and the refinement of professional judgment over time. The study therefore offers a bounded explanation of how digital engagement can support, rather than substitute for, the longer-term development of professional capital in expert-led professional communities.

Despite this study’s theoretical and practical contributions, several limitations remain. First, because the questionnaire survey was cross-sectional, it is challenging to capture dynamic changes in teachers’ training status, and reliance on self-report may introduce bias. Second, the research sample was largely drawn from a specific type of Expert Teacher Workshop in a single region, which may limit generalizability. To address these constraints, future research should examine whether reported continuance intention is subsequently enacted in practice, rather than assuming a perfect correspondence between intention and sustained participation. This can be effectively achieved through longitudinal or multi-site comparative designs. Furthermore, while this study incorporated a preliminary qualitative follow-up, future investigations could employ more extensive ethnographic observations or in-depth interviews to capture the multi-dimensional evolution of teachers’ training experiences over time. Finally, integrating additional external support elements and moderating factors, such as teachers’ technology adaptability and subject backgrounds, would further validate the TAM-ECT approach across diverse educational contexts. Through continued expansion, the analytical framework proposed here holds promise for providing solid theoretical foundations and actionable guidance to broader online professional development initiatives.

Data availability statement

All data presented in this table were collected by the author. The data that support the findings of this study are available from the corresponding author upon reasonable request.

Author contributions

Haojun Li and Yuying Luo contributed equally and share first authorship.

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